



Practical-8

Amazon Food Review

```
import sys
from itertools import combinations
from pyspark.sql import SparkSession
```

```
def main(input_path):
```

```
    spark = SparkSession.builder.appName("Amazon Food  
        Paris").getOrCreate()
    sc = spark.sparkContext
```

```
    df = spark.read.option("header", "true").csv
```

```
        (input_path)
    user_product = df.select("UserId", "ProductId")
    rdd =
```

```
        .map(lambda row: (row["UserId"],
                           row["ProductId"]))
        .filter(lambda x: x[0] is not None and
                           x[1] is not None)
```

```
    user_items = user_product.groupByKey().map
        Values (get)
```

```
    item_pairs = user_items.flatMap (
```

```
        lambda x: [(a, b), (b, a)] for a, b in
        combinations(sorted(x[1], 2))
    )
```



```
for (item1, item2), count in top20:  
    print (f"({item1}, {item2}), {count}")  
spark.stop()
```

```
if __name__ == "__main__":
```

```
    if len(sys.argv) != 2:  
        print ("Usage: amazon-item-pairs.py  
              <input-file>")
```

```
    sys.exit(-1)  
main(sys.argv[1])
```